

C. Mixed Training

time limit per test: 2 seconds

memory limit per test: 1024 megabytes



Goku wants to train for the next Tenkaichi Budokai tournament that will take place in about a month. He decided to learn some new power enhancing techniques during his free time. But as he currently has a day job, he has a limited amount of m minutes available before the tournament starts.

There are n total techniques that Goku can master, numbered 1 to n . The i -th technique is defined by three integer parameters:

- t_i : the number of minutes the technique requires.
- h_i : the amount by which Goku may increase his power level if he masters this technique.
- c_i : the *kind* of technique, an integer between 1 and k , where k is the total number of *kinds* of techniques.

Goku asked Master Roshi for his advice about the benefits of learning as many different *kinds* of techniques as possible: it turns out that each additional *kind* of technique boosts his power level by b extra points. Two techniques i and j are considered of different kinds if $c_i \neq c_j$. So, if Goku learns x different *kinds* of techniques, he earns $(x - 1) \cdot b$ extra points.

Determine the maximum power level points Goku can gain by mastering an optimal subset of techniques.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10$). The description of the test cases follows.

The first line of each test case contains four integers n ($1 \leq n \leq 1000$), k ($1 \leq k \leq n$), m ($1 \leq m \leq 10^4$), and b ($1 \leq b \leq 10^4$) — n is the number of available techniques, k is the number of *kinds* of techniques, m is the number of minutes Goku has available, and b is the number of bonus points per each additional distinct technique Goku learns.

Each of the next n lines describes a technique. The i -th of these lines contains three integers t_i ($1 \leq t_i \leq 10^4$), h_i ($1 \leq h_i \leq 10^4$), and c_i ($1 \leq c_i \leq k$) — the number of minutes this technique requires, the number of power level points this technique provides, and the kind of technique, respectively.

It is guaranteed that the sum of n over all test cases does not exceed 1000.

Output

For each test case, output a single integer — the maximum possible power level points Goku can gain.

Scoring

This problem is divided into the following subtasks.

Subtask	Points	Additional Restrictions
0	0	Example test cases
1	12	$n \leq 15$

2	19	$t_i = 10$ for $1 \leq i \leq n$
3	16	$k = 1$
4	32	$k \leq 10$, $n \leq 100$, and $m \leq 1000$
5	21	No additional restrictions

Example

input

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```

3
4 3 10 6
1 15 2
4 20 1
6 30 1
8 32 3
1 1 9 10000
10 1 1
5 1 15 1
12 4 1
2 2 1
1 2 1
1 1 1
4 10 1

```

output

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```

53
0
15

```

Note

In the first sample test case, an optimal way is to learn the first and last techniques, where Goku spends $1 + 8 = 9$ minutes and obtains $15 + 32 + 6 = 53$ power level points. The 6 points are gained as bonus because Goku learns two different kinds of techniques.